INCH-POUND

MIL-PRF-1/878G
w/AMENDMENT1
12 February 2018
SUPERSEDING
MIL-PRF-1/878G
27 March 2012

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 250TH *

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Triode.

See figure 1.

Weight: 10 ounces nominal (283.5 grams).

Mounting position: Vertical.

ABSOLUTE RATINGS: C Telegraphy.

Parameter:	F	Ef	Eb	Ec	Ib	Ic	Pp
Unit:	MHz	V	kV dc	V dc	mA dc	mA dc	W
Maximum:	40	5.25	3.0	-210	330	75.0	250
Minimum:	---	4.75	---	---	---	---	---
Test conditions:	---	5.0 V ac	2.5	Adjust	100	---	---

See footnotes at end of table I.

GENERAL:

Qualification: Not required.

Holding period: (MIL-STD-1311): 1/

This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.

* Tube type 250TL has been deleted from this tube specification sheet. Use EIA Type 250TL.

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TABLE I. Testing and inspection.

Inspection	Method MIL-STD-1311	Conditions	Symbol	Limits		Unit
				Min	Max	
Conformance inspection, part 1						
Filament current	1301		If	9.7	11.2	A ac
Total grid current	1266		Ic	---	-15	μA dc
Electrode voltage (grid)	1261		Ec	-27.0	-39.0	V dc
Primary grid emission	1266	Ef = 5.5 V ac; Ic = 175 mA dc; t = 15	Isg	---	-500	μA dc
Peak emission	1231	eb = ec = 2,500 v	is	5.0	---	a
Conformance inspection, part 2						
Amplification factor	1316		Mu	31.0	41.0	---
Power oscillation	1236	Eb = 3,000 V dc; Ib = 300 mA dc; F = 8 MHz	Po	540	---	W
Direct-interelectrode capacitance	1331		Cgp Cin Cout	2.2 3.7 ---	3.0 5.1 0.6	pF pF pF

See footnotes at end of table.

TABLE I. Testing and inspection - Continued.

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Inspection	Method MIL-STD-1311	Conditions	Symbol	Limits		Unit
				Min	Max	
Conformance inspection, part 3						
Life test	---	Group C; power oscillation; t = 500 hours	---	---	---	---
Life-test end points:	---					
Peak emission	1231		is	3.5	---	a
Primary grid emission	1266		Isg	---	-500	μA dc
Low-frequency vibration	1031	No voltages 2/	---	---	---	---
Bump	1036	Angle = 20° 2/	---	---	---	---
Low-frequency vibration and bump end points:	---					
Total grid current	1266		Ic	---	-15	μA dc
Electrode voltage (grid)	1261		Ec	-27.0	-39.0	V dc

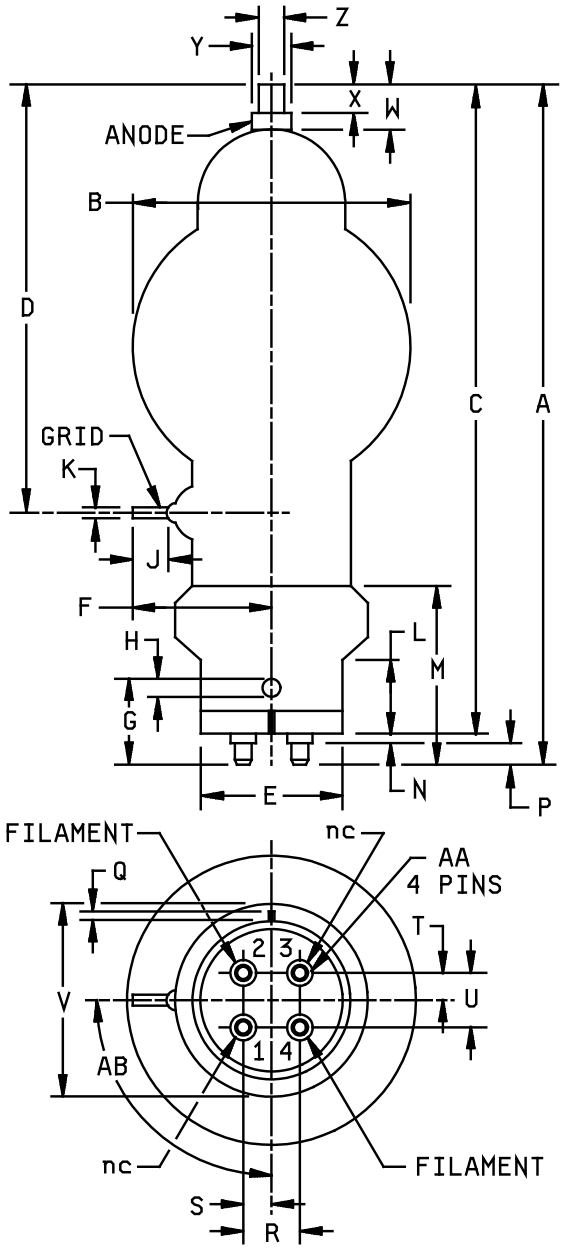
1/ In addition to the 24-hour holding period, 10 percent of the tubes in any production lot shall be held for a period of 72 hours. If more than 2 percent of the tubes fail to meet the total grid current test, the tubes represented by this sample shall be subject to recall and all tubes shall be held for a period of 72 hours.

2/ This test shall be performed yearly, with sampling as follows:

n1 = 4	c1 = 0	where c2 represents the total failures for the
n2 = 4	c2 = 0;	first and second samples combined.

In case of failure, the test shall become a conformance inspection, part 2 test for three successive successful submissions, at which time the test may revert to the yearly basis.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
Conformance inspection, part 2					
A	9.625	10.125	244.48	257.18	
B	---	3.813	---	96.85	
D	5.688	6.188	144.48	157.18	
F	1.563	1.938	39.70	49.23	
J	.312	---	7.92	---	
Conformance inspection (periodic check) part 3					1
C	9.375	9.875	238.13	250.83	
E	---	1.867	---	47.42	
H	---	.082	---	2.08	
K	.065	.071	1.65	1.80	
P	.250	.320	6.35	8.13	2
Q	---	.109	---	2.77	
V	---	2.500	---	63.50	
X	.328	---	8.33	---	
Z	.350	.365	8.89	9.27	
AA	.185	.189	4.70	4.80	
AB	80°		100°		
Reference dimensions					
G	1.225		31.12		2
L	1.938		49.23		
M	2.719		69.06		2
N	.031		0.79		
R	.688		17.48		
S	.344		8.74		
T	.344		8.74		
U	.688		17.48		
W	.563		14.30		
Y	.500		12.70		

NOTES: 1. This test shall be performed yearly, with sampling as follows:

n1 = 4	c1 = 0	where c2 represents the total failures for the
n2 = 4	c2 = 0;	first and second samples combined.

In case of failure, the test shall become a conformance inspection, part 2 test for three successive successful submissions, at which time the test may revert to the yearly basis.

2. On finished tube, add .060 inch (1.52 mm) maximum for solder.

FIGURE 1. Outline drawing of electron tube type 250TH.

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Referenced documents. In addition to MIL-PRF-1, this specification sheet references MIL-STD-1311.

The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2018-007)

Review activities:
Army - AR
Navy - AS, CG, MC, OS, SH
Air Force - 19, 99

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